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Preface

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List of Notations

The following table lists the notation used throughout this report, organized by topic.

Symbol	Meaning
General	
$\mathbb{N}$	Set of all natural numbers i.e., positive integers
$\mathbb{Z}$	Set of all integers
$\mathbb{Z}_0$	Set of all non-negative integers
$\mathbb{R}$	Set of real numbers
$\mathbb{C}$	Set of complex numbers
$[n]$	The set $\{1, 2, \ldots, n\}$ for $n \in \mathbb{N}$
i	Imaginary unit: $i^2 = -1$
S^1	$\{z \in \mathbb{C} : \lvert z\rvert = 1\}$, the multiplicative group of unit complex numbers
ζ_m	A primitive m-th root of unity
$[x]$	Floor function: largest integer $\leq x$; also written $\lfloor x \rfloor$
Finite Fields	
$\mathbb{F}_q$	Finite field with q elements
$\mathbb{F}_q^*$	Multiplicative group of nonzero elements of $\mathbb{F}_q$
$\mathbb{F}_p$	Prime field with p elements, $p = \mathrm{char}(\mathbb{F}_q)$
$\mathbb{F}_q^{*\,(d)}$	Subgroup of d-th powers in $\mathbb{F}_q^*$
E, F, L, K	Generic notation for finite fields or finite extensions of $\mathbb{F}_q$
$[E : \mathbb{F}_q]$	Degree of the extension $E/\mathbb{F}_q$
$\mathbb{F}_q[X_1, \ldots, X_n]$	Polynomial ring in n variables over $\mathbb{F}_q$
$\mathbb{F}_q^{\leq d}[X_1, \ldots, X_n]$	Polynomials in $\mathbb{F}_q[X_1, \ldots, X_n]$ of total degree at most d
Characters of Finite Abelian Groups	
G	A finite Abelian group with identity element 1_G
Z_k	Cyclic multiplicative group of order k
χ_0	Trivial character of G: $\chi_0(g) = 1$ for all $g \in G$
$\overline{\chi}$	Conjugate (and inverse) character of χ: $\overline{\chi}(g) = \overline{\chi(g)}$
$\mathrm{Hom}(A, B)$	Set of all group homomorphisms from A to B
$\widehat{G} = \mathrm{Hom}(G, S^1)$	Dual group of G; the group of all characters of G
$\mathrm{Ann}(\chi)$	Annihilator of character χ: $\{g \in G : \chi(g) = 1\}$, a subgroup of G
$\mathrm{Ann}(H)$	Annihilator of subgroup $H \leq G$: $\{\chi \in \widehat{G} : \chi(h) = 1\ \forall h \in H\}$

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Symbol	Meaning
ev_g	Evaluation character of $\widehat{G}$ defined by $\mathrm{ev}_g(\chi) = \chi(g)$
Maps on Finite Fields	
$\mathrm{Tr}\colon \mathbb{F}_q \to \mathbb{F}_p$	Absolute trace function
$N\colon \mathbb{F}_q \to \mathbb{F}_p$	Absolute norm function
$\mathrm{Tr}_{E/\mathbb{F}_q}$	Trace function from E to $\mathbb{F}_q$
$\mathrm{N}_{E/\mathbb{F}_q}$	Norm function from E to $\mathbb{F}_q$
$\sigma\colon \mathbb{F}_q \to \mathbb{F}_q$	Frobenius automorphism: $\sigma(\alpha) = \alpha^p$
Additive Characters of $\mathbb{F}_q$	
χ, τ	Generic notation for additive characters
χ_1	Canonical additive character of $\mathbb{F}_q$: $\chi_1(\alpha) = e^{2\pi i\,\mathrm{Tr}(\alpha)/p}$
χ_β	Additive character defined by $\chi_\beta(\alpha) = \chi_1(\beta\alpha)$, for $\beta \in \mathbb{F}_q$
$\boldsymbol{\mathcal{X}}_q$	Set of all additive characters of $\mathbb{F}_q$
$\chi^{(r)}$	Additive character of $\mathbb{F}_q$ lifted to E via $\chi^{(r)} = \chi \circ \mathrm{Tr}_{E/\mathbb{F}_q}$, $[E : \mathbb{F}_q] = r$
Multiplicative Characters of $\mathbb{F}_q$	
ψ, λ	Generic notation for multiplicative characters
ψ_0	Trivial multiplicative character of $\mathbb{F}_q$
η	Quadratic character of $\mathbb{F}_q$ (for odd q): $\eta(\alpha) = \pm 1$ according as α is a square
$\boldsymbol{\mathcal{M}}_q$	Set of all multiplicative characters of $\mathbb{F}_q^*$
$\boldsymbol{\mathcal{M}}_q^{(e)}$	Set of multiplicative characters of $\mathbb{F}_q$ of exponent e
$\mathrm{ord}(\psi)$	Order of ψ: smallest positive integer d with $\psi^d = \psi_0$
$\left(\frac{\alpha}{q}\right)$	Legendre symbol; equals $\eta(\alpha)$ for $\alpha \in \mathbb{F}_q$
$\psi^{(r)}$	Multiplicative character of $\mathbb{F}_q$ lifted to E via $\psi^{(r)} = \psi \circ \mathrm{N}_{E/\mathbb{F}_q}$, $[E\colon \mathbb{F}_q] = r$
Gaussian and Jacobi Sums	
$G(\psi, \chi)$	Gaussian sum: $\sum_{\alpha \in \mathbb{F}_q^*} \psi(\alpha)\,\chi(\alpha)$, for $\psi \in \boldsymbol{\mathcal{M}}_q$, $\chi \in \boldsymbol{\mathcal{X}}_q$
$J_\alpha(\lambda_1, \ldots, \lambda_k)$	Generalized Jacobi sum: $\displaystyle\sum_{\substack{\alpha_1,\ldots,\alpha_k \in \mathbb{F}_q \\ \alpha_1+\cdots+\alpha_k=\alpha}} \lambda_1(\alpha_1)\cdots\lambda_k(\alpha_k)$
$J(\lambda_1, \ldots, \lambda_k)$	Jacobi sum; special case $\alpha = 1$ of J_α
$J_0(\lambda_1, \ldots, \lambda_k)$	Jacobi sum at zero; special case $\alpha = 0$ of J_α
Kloosterman Sums	
$K(\chi; a, b)$	Kloosterman sum: $\sum_{\gamma \in \mathbb{F}_q^*} \chi(a\gamma + b\gamma^{-1})$, for $\chi \in \boldsymbol{\mathcal{X}}_q$, $a, b \in \mathbb{F}_q$
$K(\chi^{(r)}; a, b)$	Kloosterman sum lifted to the degree-r extension of $\mathbb{F}_q$
w_1, w_2	Reciprocal roots of the L-function associated to a Kloosterman sum
Salié and Jacobsthal Sums	
$S(\chi, \tau)$	Salié sum: $\sum_{\gamma \in \mathbb{F}_q^*} \eta(\gamma)\,\chi(\gamma)\,\tau(\gamma^{-1})$, for $\chi, \tau \in \boldsymbol{\mathcal{X}}_q$, q odd
$H_n(\alpha)$	Jacobsthal sum: $\sum_{\gamma \in \mathbb{F}_q} \eta(\gamma^{n+1} + \alpha\gamma)$, for $\alpha \in \mathbb{F}_q^*$

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Symbol	Meaning
$I_n(\alpha)$	Associated Jacobsthal sum: $\sum_{\gamma\in\mathbb{F}_q} \eta(\gamma^n+\alpha)$, for $\alpha \in \mathbb{F}_q^*$
Classical Riemann Zeta Function	
$\zeta(s)$	Riemann zeta function: $\sum_{n=1}^{\infty} n^{-s}$, for $\Re(s) > 1$
$\xi(s)$	Completed zeta function: $\frac{1}{2}s(s-1)\,\Gamma(s/2)\,\pi^{-s/2}\,\zeta(s)$
$\Gamma(s)$	Gamma function: $\int_0^\infty e^{-t}t^{s-1}\,dt$
$\Lambda(n)$	Von Mangoldt function: $\log p$ if $n = p^k$, else 0
ρ	Non-trivial zero of $\xi(s)$
Dirichlet L-functions over $\mathbb{F}_q[X]$	
Φ	Set of all monic polynomials in $\mathbb{F}_q[X]$
Φ_d	Monic polynomials in $\mathbb{F}_q[X]$ of degree exactly d
$\mathrm{Irr}(\Phi)$	Set of irreducible monic polynomials in Φ
$\mathfrak{n}(f) = q^{\deg(f)}$	Norm of the polynomial $f \in \mathbb{F}_q[X]$
G	Group of rational functions $r = h_1/h_2$ in $\mathbb{F}_q(X)$ with $h_1, h_2 \in \Phi$ monic
$\overline{G}$	Subgroup of G on which a given character is defined; extended to G by zero
$L(s,\chi)$	Dirichlet L-function: $\sum_{h\in\Phi} \chi(h)\,\mathfrak{n}(h)^{-s}$, for a multiplicative character χ
$\overline{L}(z,\chi)$	Power series satisfying $\overline{L}(z,\chi) = \sum_{h\in\Phi} \chi(h)\,z^{\deg(h)}$ and $\overline{L}(q^{-s},\chi) = L(s,\chi)$
L_k	k-th log-derivative coefficient: $\displaystyle\sum_{\substack{h\in\mathrm{Irr}(\Phi)\\ \deg(h)\mid k}} \deg(h)\,\chi(h)^{k/\deg(h)}$
Equations over $\mathbb{F}_q$ *— Solution Sets*	
$\mathcal{V}(f)$	Solution set of $f(X_1,\ldots,X_n) = 0$ in $\mathbb{F}_q^n$
$Z(f)$	Number of solutions: $\lvert\mathcal{V}(f)\rvert$
$\overline{\mathcal{V}}(f)$	Set of *non-trivial* solutions (excluding $(0,\ldots,0)$ if it's a zero) of $f = 0$
$\overline{Z}(f)$	Number of non-trivial solutions: $\lvert\overline{\mathcal{V}}(f)\rvert$
$\mathcal{V}(f_1,\ldots,f_k)$	Common solution set: $\mathcal{V}(f_1)\cap\cdots\cap\mathcal{V}(f_k)$
$Z(f_1,\ldots,f_k)$	Count of common solutions: $\lvert\mathcal{V}(f_1,\ldots,f_k)\rvert$
$Z_E(P=0)$	Number of solutions of $P(X_1,\ldots,X_n) = 0$ in E where $P \in \mathbb{F}_q[X_1,\ldots,X_n]$, $[E\colon \mathbb{F}_q] > 1$
Ω_d	Space $\mathbb{F}_q^{\leq d}[X_1,\ldots,X_n]$ of polynomials of total degree at most d
$\omega(d)$	Set of n-tuples $(i_1,\ldots,i_n) \in \mathbb{Z}_0^n$ with $i_1+\cdots+i_n \leq d$
Quadratic Forms	
C_f	Coefficient matrix of a quadratic form f: $(C_f)_{ij} = a_{ij}$ where $f = X^\top C_f X$
$\det(f)$	Determinant of the coefficient matrix C_f
$\vartheta : \mathbb{F}_q \to \mathbb{Z}$	Auxiliary function: $\vartheta(0) = q-1$ and $\vartheta(\alpha) = -1$ for $\alpha \neq 0$
Diagonal Equations	
$d_i = \gcd(k_i, q-1)$	Exponent reduced modulo the order of $\mathbb{F}_q^*$, for $a_iX_i^{k_i}$ in a diagonal equation
$M(d_1,\ldots,d_n)$	$\#\left\{(t_1,\ldots,t_n) \in \prod_i [d_i-1] : \frac{t_1}{d_1}+\cdots+\frac{t_n}{d_n} \in \mathbb{Z}\right\}$
$D = \mathrm{lcm}(d_1,\ldots,d_n)$	LCM of exponents; appears in the closed-form formula for $M(d_1,\ldots,d_n)$
Character Sums with Polynomials (Weil Sums)	

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Symbol	**Meaning**
$W(\chi; f)$	Additive Weil sum: $\sum_{\alpha \in \mathbb{F}_q} \chi(f(\alpha))$, for $\chi \in \boldsymbol{\mathcal{X}}_q$, $f \in \mathbb{F}_q[X]$
$W(\psi; f)$	Multiplicative Weil sum: $\sum_{\alpha \in \mathbb{F}_q} \psi(f(\alpha))$, for $\psi \in \boldsymbol{\mathcal{M}}_q$, $f \in \mathbb{F}_q[X]$
$W(\psi, \chi; f, g)$	Mixed Weil sum: $\sum_{\alpha \in \mathbb{F}_q} \psi(f(\alpha))\, \chi(g(\alpha))$, for $\psi \in \boldsymbol{\mathcal{M}}_q$, $\chi \in \boldsymbol{\mathcal{X}}_q$, $f, g \in \mathbb{F}_q[X]$
Lifted Weil Sums (over Extension Fields)	
$E = \mathbb{F}_{q^k}$	Degree-k extension of $\mathbb{F}_q$; $[E : \mathbb{F}_q] = k$
$W^{(k)}(\psi^{(k)}; f)$	Lifted multiplicative Weil sum: $\sum_{\alpha \in E} \psi^{(k)}(f(\alpha))$
$W^{(k)}(\chi^{(k)}; g)$	Lifted additive Weil sum: $\sum_{\alpha \in E} \chi^{(k)}(g(\alpha))$
$W^{(k)}(\psi^{(k)}, \chi^{(k)}; f, g)$	Lifted mixed Weil sum: $\sum_{\alpha \in E} \psi^{(k)}(f(\alpha))\, \chi^{(k)}(g(\alpha))$
Special L-functions (Weil chapter)	
$\overline{G}$ (Weil context)	The specific subgroup $\{r \in G : h_1(\gamma_i) \cdot h_2(\gamma_i) \neq 0\ \forall i\}$ for roots γ_i of a fixed f
$\varsigma : \overline{G} \to \mathbb{F}_q$	Multiplicative map $\varsigma(r) = \prod_i f(\alpha_i) \cdot \prod_j f(\beta_j)^{-1}$ for $r = \prod(X - \alpha_i)/\prod(X - \beta_j)$
$\varrho : G \to \mathbb{F}_q$	Additive map $\varrho(r) = \sum_i g(\alpha_i) - \sum_j g(\beta_j)$ for a fixed polynomial g
$\xi : G \to \mathbb{C}$	Combined character: $\xi(r) = (\psi \circ \varsigma)(r) \cdot (\chi \circ \varrho)(r)$
$L(s, \xi)$	L-function on ξ: $\sum_{h \in \Phi} \xi(h)\, \mathfrak{n}(h)^{-s}$
$\overline{L}(z, \xi)$	Power series: $\sum_{h \in \Phi} \xi(h)\, z^{\deg(h)}$, related by $\overline{L}(q^{-s}, \xi) = L(s, \xi)$
$L^{(k)}(s, \xi^{(k)})$	L-function on the lifted character $\xi^{(k)}$ over $\mathbb{F}_{q^k}$
$\overline{L}^{(k)}(z, \xi^{(k)})$	Power series form of the lifted L-function
$\Phi^{(k)}$	Set of monic polynomials over $E = \mathbb{F}_{q^k}$
$\mathfrak{n}^{(k)}(h) = q^{k \cdot \deg(h)}$	Norm of $h \in \Phi^{(k)}$ over E; analog of $\mathfrak{n}(h) = q^{\deg(h)}$ for $\mathbb{F}_q$
Absolute Varieties	
$\mathcal{A}$	Variety defined over $\mathbb{F}_q$
$Z_k(\mathcal{A})$	Number of $\mathbb{F}_{q^k}$-rational points of $\mathcal{A}$: $\lvert\{(\alpha_1, \ldots, \alpha_n) \in \mathcal{A} : \alpha_i \in \mathbb{F}_{q^k}\}\rvert$

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Chapter 1

Finite Fields

§ 1.1 Field Extensions

We build up the theory of field extensions following Morandi [Mor96].

The study of field theory is primarily the study of field extensions. The classical problems of ruler-and-compass constructions and the solvability of polynomial equations by radicals were settled by analyzing appropriate field extensions. The central idea is to associate a group—the *Galois group*—to a field extension, converting field-theoretic questions into group-theoretic ones. Since the Galois group of a finite extension is finite, the rich structure theory of finite groups becomes available. In this section we build the foundational vocabulary: degrees, algebraic elements, minimal polynomials, and the key structural results about simple and algebraic extensions.

Degrees and Generated Subfields

Let $F \subseteq K$ are fields. Then K is called a *field extension* of F. We denote it by K/F. Here F is called the *base field*. We can think of K as a vector space over F where the addition of two vectors follows the addition operation of K and the multiplication of any $\alpha \in K$ and $c \in F$ follows the multiplication operation of K, $c \cdot \alpha = c\alpha$. Then we denote the dimension of K over F by

$$[K\colon F] \coloneqq \dim_F(K)$$

This dimension is called the *degree of field extension* of F or simply *degree of* $[K\colon F]$. If $[K\colon F] < \infty$ we call it by a special name, *finite extension*. Finite extensions of finite fields are of very important to us and this report we will work on finite fields and their finite extensions. Let $\alpha \in K$. Then we denote $F(\alpha)$ to be the field generated by F and α following the multiplication and addition operations of K. We will define what it means to be the field generated by F and α in Definition 1.1.1. It is very easy to sea $F(\alpha)$ is also a field, and it contains F but also contained in K.

Let F be any field and suppose X be a formal variable. Then $F[X]$ denotes the polynomial ring over F. We define the rational function field $F(X)$ to be the *fraction ring* of $F[X]$ which consists of all quotients f/g where $f, g \in F[X]$ with $g(X) \neq 0$. Similarly, we can define the polynomial ring $F[X_1, \ldots, X_n]$ over multiple formal variables and their fraction field $F(X_1, \ldots, X_n)$.

Look at the following example. $\mathbb{R} \subseteq \mathbb{C}$ are both fields. But $\mathbb{C}$ is a 2-dimensional vector space over $\mathbb{R}$ as we can take the basis $\{1, i\}$. So $\mathbb{C}/\mathbb{R}$ is a finite extension of degree 2. But $\mathbb{Q}/\mathbb{R}$ is an infinite field extension. On the other hand let $\alpha \in \mathbb{R}$. Then consider the field $\mathbb{Q}(\alpha)$. Now degree of $[\mathbb{Q}(\alpha)\colon \mathbb{Q}]$ depends on what kind of element α is. If $\alpha = \sqrt{2}$ then $[\mathbb{Q}(\sqrt{2})\colon \mathbb{Q}] = 2$ but if $\alpha = \pi$ then $[\mathbb{Q}(\pi)\colon \mathbb{Q}] = \infty$.

Definition 1.1.1: Generated Subfields

Let K/F be a field extension and $S \subseteq K$.

(i) The ring generated by F and S, *written $F[S]$, is the intersection of all subrings of K containing F and S.*

(ii) The field generated by F and S, *written $F(S)$, is the intersection of all subfields of K containing F and S.*

If $S = \{a_1, \dots, a_n\}$ is finite we write $F[a_1, \dots, a_n]$ and $F(a_1, \dots, a_n)$. We call $F(\alpha_1, \dots, \alpha_n)$ finitely generated *field.*

From this definition it easily follows that for any $S \subseteq K$, $F(S)$ is the smallest subfield of K containing F and α.

Let K/F be a field extension. For any $\alpha \in K$ consider the map $\mathrm{ev}_\alpha : F[X] \to K$ defined by $f(X) \mapsto f(\alpha)$. Hence, ev_α is a ring homomorphism.

Observation 1.1. *For any $\alpha \in K$, $\mathrm{Im}(\mathrm{ev}_\alpha)$ is a subring of K and ev_α is F-vector space homomorphism.*

Now we can use this map ev_α to give descriptions of the elements of $F[\alpha]$ and $F(\alpha)$.

Lemma 1.1.1

Let K be a field extension of F and let $a \in K$. Then

$$F[a] = \{f(a) : f(x) \in F[x]\} \quad \text{and} \quad F(a) = \left\{\frac{f(a)}{g(a)} : f, g \in F[x],\ g(a) \neq 0\right\}.$$

Moreover, $F(a)$ is the quotient field of $F[a]$.

Proof: The evaluation map ev_a is a ring homomorphism with image $\{f(a) : f \in F[x]\}/$ So this set is a subring of K. Any subring of K containing F and a must contain $f(a)$, for all $f \in F[X]$. Therefore, $F[a] = \{f(a) : f \in F[x]\}$. The fraction field of $F[a]$ is $\{f(a)/g(a) \colon g(a) \neq 0\}$; it is contained in any subfield of K containing $F[a]$, hence equals $F(a)$. ■

We call this procedure *adjoining* α to F. Using similar arguments we can describe the ring $F[a_1, \dots, a_n]$ and the field $F(a_1, \dots, a_n)$ where K/F is a field extension and $a_1, \dots, a_n \in K$. So we have the following theorem.

Theorem 1.1.2

Let K be a field extension of F and let $a_1, \dots, a_n \in K$. Then

$$F[a_1, \dots, a_n] = \{f(a_1, \dots, a_n) : f \in F[x_1, \dots, x_n]\},$$
$$F(a_1, \dots, a_n) = \left\{\frac{f(a_1, \dots, a_n)}{g(a_1, \dots, a_n)} : f, g \in F[x_1, \dots, x_n],\ g(a_1, \dots, a_n) \neq 0\right\}.$$

Moreover, $F(a_1, \dots, a_n)$ is the quotient field of $F[a_1, \dots, a_n]$.

Now we can also define fields generated by adjoining arbitrary number of elements. In this case we can give the description of the new field in terms of all finite extension of fields created by adjoining finitely many elements.

Theorem 1.1.3

Let K be a field extension of F and let $S \subseteq K$ be any subset. If $\alpha \in F(S)$, then $\alpha \in F(a_1, \dots, a_n)$ for some $a_1, \dots, a_n \in S$. Therefore,

$$F(S) = \bigcup \{F(a_1, \dots, a_n) \colon a_1, \dots, a_n \in S\},$$

where the union is over all finite subsets of S.

Proof: Let $U := \bigcup \{F(a_1, \ldots, a_n) : a_1, \ldots, a_n \in S\}$ Each $F(a_1, \ldots, a_n)$ contains F and lies in $F(S)$ where $a_1, \ldots, a_n \in S$. So the union U satisfies $U \subseteq F(S)$. The set U contains F and S. If $\alpha \in F(a_1, \ldots, a_n)$ and $\beta \in F(b_1, \ldots, b_m)$, then $\alpha \pm \beta$, $\alpha\beta$, α/β (if $\beta \neq 0$) are all in $F(a_1, \ldots, a_n, b_1, \ldots, b_m) \subseteq U$. Therefore, U is a field. Being the smallest subfield containing F and S, it equals $F(S)$. ■

Algebraic Elements and Minimal Polynomials

Definition 1.1.2: Algebraic and Transcendental Elements

Let K be a field extension of F. An element $\alpha \in K$ is algebraic over F *if there is a nonzero polynomial $f(x) \in F[x]$ with $f(\alpha) = 0$. If α is not algebraic over F, then α is* transcendental over F. *If every element of K is algebraic over F, then K is said to be* algebraic over F, *and K/F is called an* algebraic extension.

Hermite (1873) proved that e is transcendental over $\mathbb{Q}$ and Lindemann (1882) proved the same for π. On the other hand, $\sqrt[n]{r} \in \mathbb{R}$ is algebraic over $\mathbb{Q}$ for any $r \in \mathbb{Q}$ (root of $x^n - r$), as is $i = \sqrt{-1}$ (root of $x^2 + 1$).

Definition 1.1.3: Minimal Polynomial

If α is algebraic over a field F, the minimal polynomial *of α over F is the monic polynomial $p(x)$ of least degree in $F[x]$ for which $p(\alpha) = 0$; it is denoted $\min(F, \alpha)$. Equivalently, $\min(F, \alpha)$ is the monic generator of the kernel of the evaluation homomorphism ev_α.*

The minimal polynomial depends on the base field. For instance, $\min(\mathbb{Q}, i) = x^2 + 1$ while $\min(\mathbb{C}, i) = x - i$, since $i \in \mathbb{C}$ is already in the base field.

Structure of Simple Algebraic Extensions

Theorem 1.1.4

Let K be a field extension of F and let $\alpha \in K$ be algebraic over F. If $p(x) = \min(F, \alpha)$, then:

(i) *$p(x)$ is irreducible over F.*

(ii) *If $g(x) \in F[x]$, then $g(\alpha) = 0$ if and only if $p(x) \mid g(x)$.*

(iii) *If $n = \deg(\min(F, \alpha))$, then the elements $1, \alpha, \ldots, \alpha^{n-1}$ form a basis for $F(\alpha)$ over F, so $[F(\alpha) : F] = \deg(\min(F, \alpha)) < \infty$. Moreover, $F(\alpha) = F[\alpha]$.*

(iv) *$F(\alpha) \cong F[x]/(p(x))$.*

Proof: The first isomorphism theorem gives $F[\alpha] \cong F[x]/(p(x))$. Since $F[\alpha] \subseteq K$ is an integral domain, $(p(x))$ is a prime ideal; in a PID every nonzero prime is maximal, so $F[x]/(p(x))$ is a field and $F[\alpha] = F(\alpha)$. This gives (i) and (iv). Part (ii) follows from $\ker(\mathrm{ev}_\alpha) = (p(x))$.

For (iii): any $b \in F(\alpha) = F[\alpha]$ satisfies $b = g(\alpha)$; dividing gives $g = qp + r$ with $\deg r < n$, so $b = r(\alpha)$ lies in the span of $\{1, \alpha, \ldots, \alpha^{n-1}\}$. A nontrivial F-linear relation among these would give a polynomial of degree $< n$ in $\ker(\mathrm{ev}_\alpha)$, contradicting minimality of p. ■

Example 1.1.1

(i) $\alpha = \sqrt[3]{2}$: the polynomial $x^3 - 2$ is irreducible over $\mathbb{Q}$ by Eisenstein at $p = 2$, so $\min(\mathbb{Q}, \sqrt[3]{2}) = x^3 - 2$ and $[\mathbb{Q}(\sqrt[3]{2}) : \mathbb{Q}] = 3$, with $\mathbb{Q}$-basis $\{1, \sqrt[3]{2}, \sqrt[3]{4}\}$.

(ii) $\omega = e^{2\pi i/p}$, p prime: since $\omega^p = 1$ and $\omega \neq 1$, it is a root of the cyclotomic polynomial $\Phi_p(x) = x^{p-1} + \cdots + x + 1$, which is irreducible over $\mathbb{Q}$. Hence $\min(\mathbb{Q}, \omega) = \Phi_p(x)$ and $[\mathbb{Q}(\omega) : \mathbb{Q}] = p - 1$.

Finite and Algebraic Extensions

Lemma 1.1.5

Let $K = F(X)$ be the field of rational functions in X over F. If $u \in K$ with $u \notin F$ let $u = f/g$ where $f, g \in F[X]$ and $\gcd(f, g) = 1$. Let $L = F(u)$. Then

$$[K : L] = \max\{\deg(f), \deg(g)\}$$

Lemma 1.1.6

If $[K : F] < \infty$, then K is algebraic and finitely generated over F.

Proof: Let $\{\alpha_1, \ldots, \alpha_n\}$ be an F-basis of K; then $K = F(\alpha_1, \ldots, \alpha_n)$, so K is finitely generated. For any $\alpha \in K$, the $n+1$ elements $1, \alpha, \ldots, \alpha^n$ span only an n-dimensional space, so they are F-linearly dependent. A nontrivial relation gives a nonzero polynomial of degree $\leq n$ in $F[x]$ vanishing at α, so α is algebraic over F. ■

Theorem 1.1.7 Tower Law

Let $F \subseteq L \subseteq K$ be fields. Then $[K : F] = [K : L] \cdot [L : F]$.

Proof: Let $\{a_i\}_{i \in I}$ be an F-basis of L and $\{b_j\}_{j \in J}$ an L-basis of K. We show $\{a_i b_j\}_{I \times J}$ is an F-basis of K.

Spanning. For $x \in K$, write $x = \sum_j \alpha_j b_j$ with $\alpha_j \in L$, and $\alpha_j = \sum_i \beta_{ij} a_i$ with $\beta_{ij} \in F$; then $x = \sum_{i,j} \beta_{ij} a_i b_j$.

Independence. If $\sum_{i,j} \beta_{ij} a_i b_j = 0$, regroup to get $\sum_j \left(\sum_i \beta_{ij} a_i\right) b_j = 0$. By L-independence of $\{b_j\}$, each $\sum_i \beta_{ij} a_i = 0$, and by F-independence of $\{a_i\}$, each $\beta_{ij} = 0$. ■

Theorem 1.1.8

Let K/F be a field extension. If $\alpha_1, \ldots, \alpha_n \in K$ are algebraic over F, then

$$[F(\alpha_1, \ldots, \alpha_n) : F] \leq \prod_{i=1}^{n} [F(\alpha_i) : F].$$

Proof: By induction on n; the case $n = 1$ is trivial. Set $L = F(\alpha_1, \ldots, \alpha_{n-1})$; by induction $[L : F] \leq \prod_{i=1}^{n-1} [F(\alpha_i) : F] < \infty$. Since α_n is algebraic over F and hence over L, with $\min(L, \alpha_n) \mid \min(F, \alpha_n)$ in $L[x]$, we have $[L(\alpha_n) : L] \leq [F(\alpha_n) : F]$. The tower law gives

$$[F(\alpha_1, \ldots, \alpha_n) : F] = [L(\alpha_n) : L] \cdot [L : F] \leq [F(\alpha_n) : F] \cdot \prod_{i=1}^{n-1} [F(\alpha_i) : F].$$

■

Corollary 1.1.9

Let K/F be a field extension and $\alpha \in K$. Then α is algebraic over F if and only if $[F(\alpha) : F] < \infty$. Moreover, K is algebraic over F if $[K : F] < \infty$.

Proof: If α is algebraic, then $[F(\alpha) : F] = \deg \min(F, \alpha) < \infty$ by Theorem 1.1.4(iii). Conversely, if $[F(\alpha) : F] < \infty$, then α is algebraic by Theorem 1.1.6. The final statement is Theorem 1.1.6. ■

Theorem 1.1.10

Let K be a field extension of F and $X \subseteq K$ such that each element of X is algebraic over F. Then $F(X)/F$ is algebraic. If $|X| < \infty$, then $[F(X) : F] < \infty$.

Proof: For any $\alpha \in F(X)$, Theorem 1.1.3 gives $\alpha \in F(a_1, \ldots, a_m)$ for some $a_1, \ldots, a_m \in X$. By Theorem 1.1.8, $[F(a_1, \ldots, a_m) : F] < \infty$, so α is algebraic over F by Theorem 1.1.6. If X is finite, the same bound gives $[F(X) : F] < \infty$. ■

Theorem 1.1.11 Transitivity of Algebraic Extensions

Let $F \subseteq L \subseteq K$ be fields. If L/F and K/L are algebraic, then K/F is algebraic.

Proof: Let $\alpha \in K$. Since K/L is algebraic, there is a nonzero $p(x) = \sum_{k=0}^{m} a_k x^k \in L[x]$ with $p(\alpha) = 0$. Each a_k is algebraic over F, so by Theorem 1.1.8 the extension $L' = F(a_0, \ldots, a_m)$ satisfies $[L' : F] < \infty$. Since $p(x) \in L'[x]$ and $p(\alpha) = 0$, the element α is algebraic over L' and $[L'(\alpha) : L'] < \infty$. By the tower law,

$$[F(a_0, \ldots, a_m, \alpha) : F] = [L'(\alpha) : L'] \cdot [L' : F] < \infty,$$

so α is algebraic over F by Theorem 1.1.6. ■

Algebraic Closure and Transitivity

Definition 1.1.4: Algebraic Closure in K

Let K be a field extension of F. The set

$$\{\, a \in K : a \text{ is algebraic over } F \,\}$$

is called the algebraic closure of F in K.

Corollary 1.1.12

Let K be a field extension of F and let L be the algebraic closure of F in K. Then L is a field, and therefore is the largest algebraic extension of F contained in K.

Proof: Let $a, b \in L$; both are algebraic over F. By Theorem 1.1.8, $F(a,b)/F$ is finite, hence algebraic by Theorem 1.1.6. Since $a \pm b$, ab, a/b $(b \neq 0)$ all lie in $F(a,b) \subseteq L$, the set L is closed under field operations, so it is a field. It is algebraic over F by definition, and it contains any algebraic extension of F inside K. ■

§ 1.2 Trace and Norm

Chapter 2

Bibliography

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